

```
# get datatypes in df
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264835 entries, 0 to 264834
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   DATE                   264835 non-null  datetime64[ns]
1   STORE_NBR              264835 non-null  int64
2   LYLTY_CARD_NBR         264835 non-null  int64
3   TXN_ID                 264835 non-null  int64
4   PROD_NBR               264835 non-null  object
5   PROD_NAME              264835 non-null  object
6   PROD_QTY               264835 non-null  int64
7   TOT_SALES              264835 non-null  float64
8   prod_name_lower        264835 non-null  object
9   ends_with_weight       264835 non-null  bool
10  WEIGHT                 264835 non-null  int64
11  LIFESTAGE              264835 non-null  object
12  PREMIUM_CUSTOMER       264835 non-null  object
13  BRAND                  264835 non-null  object
dtypes: bool(1), datetime64[ns](1), float64(1), int64(5), object(6)
memory usage: 26.5+ MB
```

```
import pandas as pd
import numpy as np

df['YEARMONTH'] = df['DATE'].dt.strftime('%Y%m')

store_monthly = (
    df.groupby(['STORE_NBR', 'YEARMONTH'])
      .agg(
          TOT_SALES=('TOT_SALES', 'sum'),
          N_CUSTOMERS=('LYLTY_CARD_NBR', 'nunique'),
          N_TXNS=('TXN_ID', 'nunique')
      )
      .reset_index()
)

store_monthly['TXN_PER_CUSTOMER'] = (
    store_monthly['N_TXNS'] /
    store_monthly['N_CUSTOMERS']
)

print(store_monthly.head())
print(store_monthly.shape)
```

	STORE_NBR	YEARMONTH	TOT_SALES	N_CUSTOMERS	N_TXNS
TXN_PER_CUSTOMER					
0	1	201807	206.9	49	52
1.061224					
1	1	201808	176.1	42	43
1.023810					
2	1	201809	278.8	59	62
1.050847					
3	1	201810	188.1	44	45
1.022727					
4	1	201811	192.6	46	47
1.021739					
(3169, 6)					

```
store_counts = (
    store_monthly.groupby('STORE_NBR')['YEARMONTH']
    .nunique()
    .sort_values()
)
```

```
print(store_counts.value_counts())
```

YEARMONTH

12 260

2 5

1 3

11 3

3 1

Name: count, dtype: int64

```
pretrial = store_monthly[
    store_monthly['YEARMONTH'] < '201902'
].copy()
```

```
print(pretrial['YEARMONTH'].unique())
```

```
['201807' '201808' '201809' '201810' '201811' '201812' '201901']
```

```
def calculate_correlation(input_table,
                           metric_col,
                           trial_store):
```

```
    trial_data = input_table[
        input_table['STORE_NBR'] == trial_store
    ][['YEARMONTH', metric_col]]
```

```
    correlations = []
```

```
    for store in input_table['STORE_NBR'].unique():
```

```
        if store == trial_store:
```

```

        continue

    control = input_table[
        input_table['STORE_NBR'] == store
    ][['YEARMONTH', metric_col]]

    merged = trial_data.merge(
        control,
        on='YEARMONTH',
        suffixes=('_trial', '_control')
    )

    corr = merged[
        f'{metric_col}_trial'
    ].corr(
        merged[f'{metric_col}_control']
    )

    correlations.append(
        [trial_store, store, corr]
    )

return pd.DataFrame(
    correlations,
    columns=[
        'TRIAL_STORE',
        'CONTROL_STORE',
        'CORRELATION'
    ]
)

def calculate_magnitude_distance(
    input_table,
    metric_col,
    trial_store):

    trial = input_table[
        input_table['STORE_NBR']==trial_store
    ][['YEARMONTH', metric_col]]

    scores = []

    for store in input_table['STORE_NBR'].unique():

        if store == trial_store:
            continue

        control = input_table[
            input_table['STORE_NBR']==store
        ][['YEARMONTH', metric_col]]

```

```

merged = trial.merge(
    control,
    on='YEARMONTH',
    suffixes=('_trial','_control')
)

distance = abs(
    merged[f'{metric_col}_trial']
    -
    merged[f'{metric_col}_control']
)

score = 1 - (
    (distance - distance.min()) /
    (distance.max() - distance.min() + 1e-9)
)

scores.append([
    trial_store,
    store,
    score.mean()
])

return pd.DataFrame(
    scores,
    columns=[
        'TRIAL_STORE',
        'CONTROL_STORE',
        'MAG_SCORE'
    ]
)

trial_stores = [77,86,88]
for trial in trial_stores:
    sales_corr = calculate_correlation(
        pretrial,
        'TOT_SALES',
        trial
    )

    sales_mag = calculate_magnitude_distance(
        pretrial,
        'TOT_SALES',
        trial
    )

    customer_corr = calculate_correlation(
        pretrial,

```

```

        'N_CUSTOMERS',
        trial
    )

customer_mag = calculate_magnitude_distance(
    pretrial,
    'N_CUSTOMERS',
    trial
)

combined = (
    sales_corr
    .merge(
        sales_mag,
        on=['TRIAL_STORE', 'CONTROL_STORE']
    )
    .merge(
        customer_corr.rename(
            columns={
                'CORRELATION':
                'CUSTOMER_CORR'
            }
        ),
        on=['TRIAL_STORE', 'CONTROL_STORE']
    )
    .merge(
        customer_mag.rename(
            columns={
                'MAG_SCORE':
                'CUSTOMER_MAG'
            }
        ),
        on=['TRIAL_STORE', 'CONTROL_STORE']
    )
)

combined['FINAL_SCORE'] = (
    combined['CORRELATION']
    +
    combined['MAG_SCORE']
    +
    combined['CUSTOMER_CORR']
    +
    combined['CUSTOMER_MAG']
) / 4

print(f"\nTRIAL STORE {trial}")
print(
    combined
    .sort_values(

```

```

        'FINAL_SCORE',
        ascending=False
    )
    .head(10)
)

```

```

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```

TRIAL	STORE 77	CONTROL_STORE	CORRELATION	MAG_SCORE	CUSTOMER_CORR
230	77	233	0.903774	0.490806	0.990358
70	77	71	0.914106	0.664887	0.754817
82	77	84	0.684348	0.629596	0.858571
116	77	119	0.867664	0.404428	0.983267
112	77	115	0.689159	0.727536	0.718882
49	77	50	0.763866	0.630361	0.607391
2	77	3	0.806644	0.522914	0.834207
40	77	41	0.783232	0.491685	0.844219
262	77	265	0.639759	0.678571	0.573460
16	77	17	0.842668	0.409456	0.747308

	CUSTOMER_MAG	FINAL_SCORE
230	0.642857	0.756949
70	0.571429	0.726310
82	0.571429	0.685986
116	0.476190	0.682887
112	0.571429	0.676751
49	0.660714	0.665583

2	0.489796	0.663390
40	0.514286	0.658356
262	0.705882	0.649418
16	0.571429	0.642715

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TRIAL STORE 86					
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152	86	155	0.877882	0.738330	0.942876
106	86	109	0.788300	0.605940	0.770778
111	86	114	0.734415	0.528371	0.855339
222	86	225	0.617533	0.586150	0.733791
135	86	138	0.759864	0.426729	0.749701
237	86	240	0.825066	0.409573	0.576403
178	86	181	0.607657	0.571478	0.549331
257	86	260	0.720350	0.397129	0.846502
219	86	222	0.795075	0.541590	0.458238
191	86	194	0.476578	0.684092	0.559963

	CUSTOMER_MAG	FINAL_SCORE
152	0.523810	0.770724
106	0.714286	0.719826
111	0.471429	0.647389
222	0.634921	0.643099
135	0.571429	0.626930
237	0.610390	0.605358
178	0.673469	0.600484

257	0.428571	0.598138
219	0.591837	0.596685
191	0.595238	0.578968

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divide
  c /= stddev[:, None]
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\
function_base.py:2898: RuntimeWarning: invalid value encountered in
divide
  c /= stddev[None, :]
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\
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  c /= stddev[:, None]
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```

TRIAL STORE 88

	TRIAL_STORE	CONTROL_STORE	CORRELATION	MAG_SCORE	CUSTOMER_CORR
175	88	178	0.731857	0.569749	0.939466
13	88	14	0.698557	0.593571	0.942976
131	88	134	0.864293	0.685875	0.508880
234	88	237	0.308479	0.595469	0.947326
184	88	187	0.599076	0.634428	0.671264
250	88	253	0.811838	0.521990	0.500962
34	88	35	0.150801	0.654197	0.899594
201	88	204	0.885774	0.509249	0.550263
68	88	69	0.450029	0.464062	0.815792
40	88	41	0.290319	0.562729	0.721650

	CUSTOMER_MAG	FINAL_SCORE
175	0.476190	0.679316
13	0.428571	0.665919
131	0.571429	0.657619
234	0.607143	0.614604
184	0.517857	0.605656
250	0.540816	0.593902
34	0.666667	0.592814
201	0.376623	0.580477

68	0.542857	0.568185
40	0.650794	0.556373

```
trial_store = 77
control_store = 233 # replace

plot_df = store_monthly[
    store_monthly['STORE_NBR'].isin(
        [trial_store, control_store]
    )
]

import matplotlib.pyplot as plt

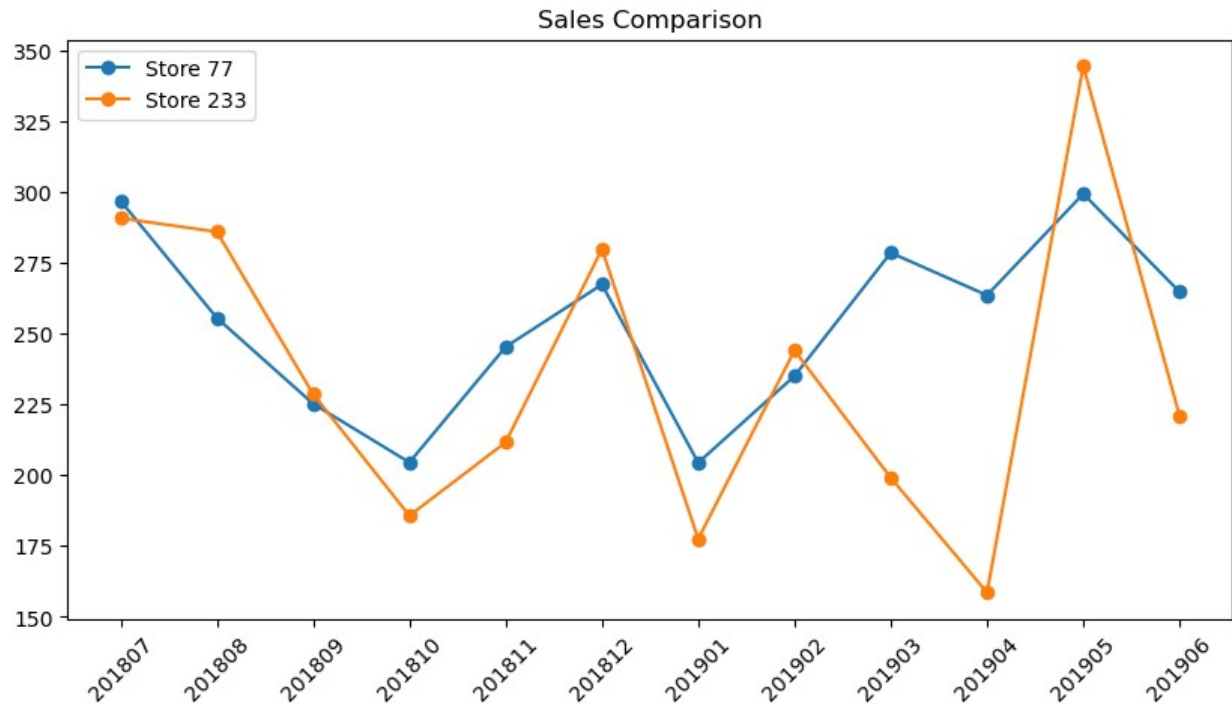
plt.figure(figsize=(10,5))

for store in [trial_store, control_store]:

    subset = plot_df[
        plot_df['STORE_NBR']==store
    ]

    plt.plot(
        subset['YEARMONTH'],
        subset['TOT_SALES'],
        marker='o',
        label=f'Store {store}'
    )

plt.legend()
plt.xticks(rotation=45)
plt.title("Sales Comparison")
plt.show()
```



```

trial_store = 86
control_store = 155 # replace

plot_df = store_monthly[
    store_monthly['STORE_NBR'].isin(
        [trial_store, control_store]
    )
]

import matplotlib.pyplot as plt
plt.figure(figsize=(10,5))

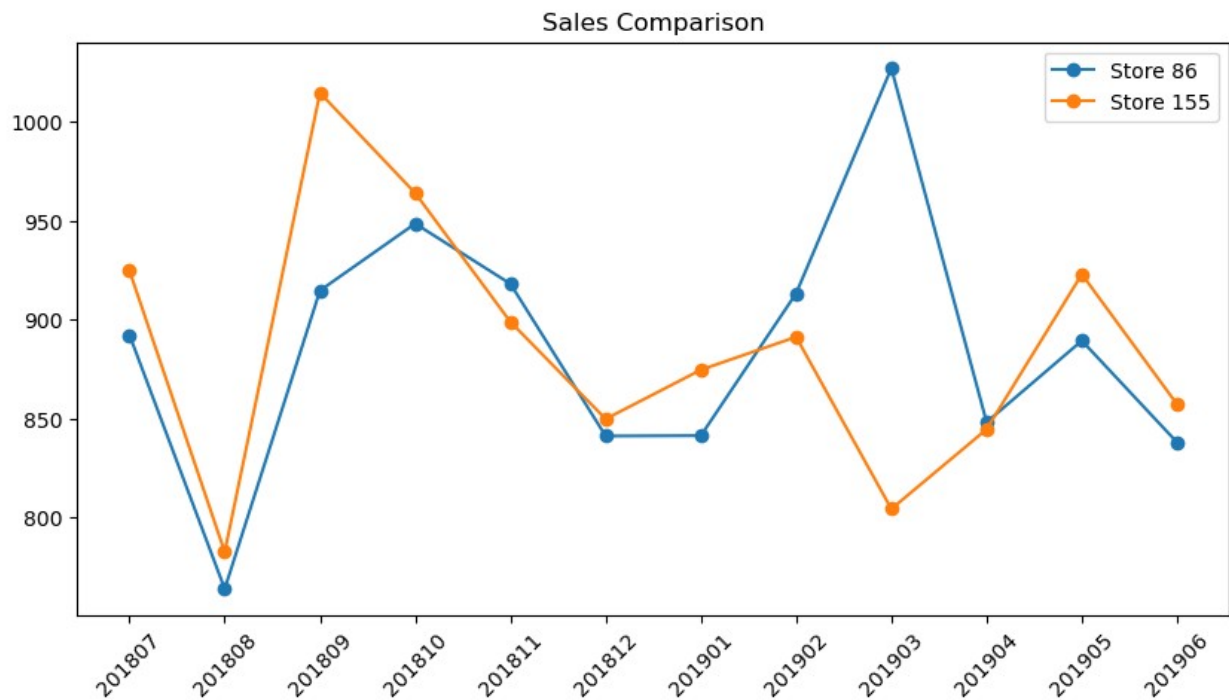
for store in [trial_store, control_store]:
    subset = plot_df[
        plot_df['STORE_NBR']==store
    ]

    plt.plot(
        subset['YEARMONTH'],
        subset['TOT_SALES'],
        marker='o',
        label=f'Store {store}'
    )

plt.legend()
plt.xticks(rotation=45)

```

```
plt.title("Sales Comparison")
plt.show()
```



```
trial_store = 88
control_store = 178 # replace

plot_df = store_monthly[
    store_monthly['STORE_NBR'].isin(
        [trial_store, control_store]
    )
]

import matplotlib.pyplot as plt

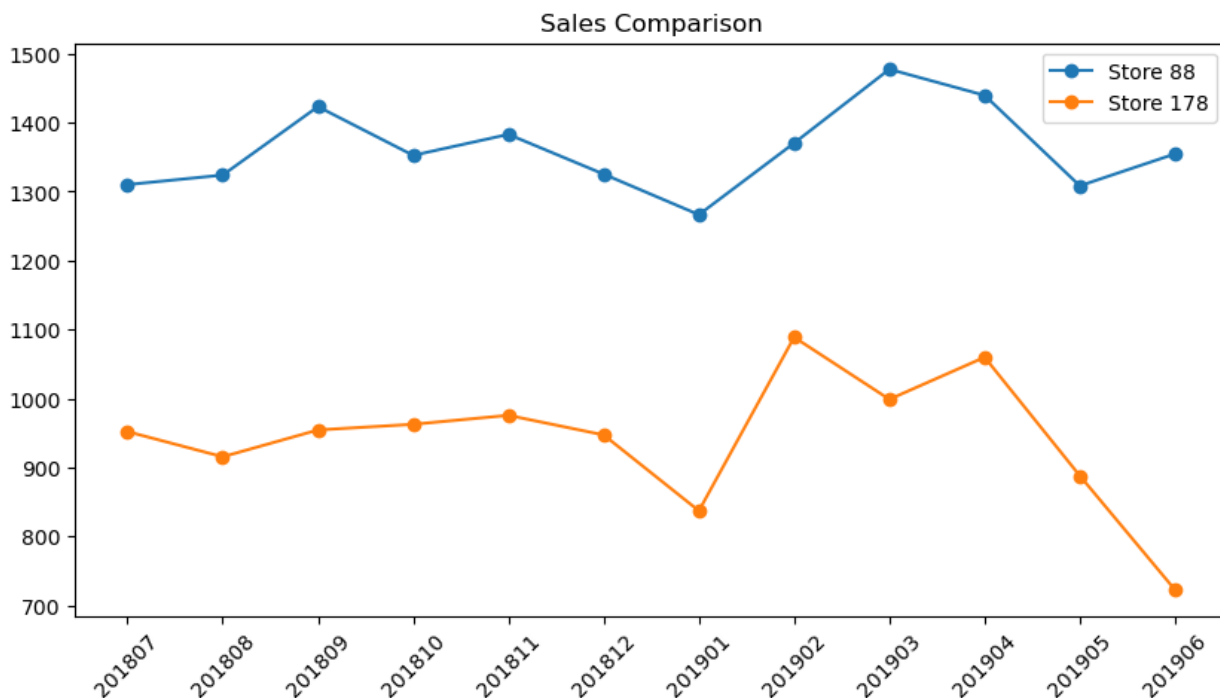
plt.figure(figsize=(10,5))

for store in [trial_store, control_store]:

    subset = plot_df[
        plot_df['STORE_NBR']==store
    ]

    plt.plot(
        subset['YEARMONTH'],
        subset['TOT_SALES'],
        marker='o',
        label=f'Store {store}'
    )
```

```
plt.legend()
plt.xticks(rotation=45)
plt.title("Sales Comparison")
plt.show()
```



```
from scipy.stats import ttest_ind

trial_period = store_monthly[
    store_monthly['YEARMONTH'] >= '201902'
]

trial_sales = trial_period[
    trial_period['STORE_NBR']==77
]['TOT_SALES']

control_sales = trial_period[
    trial_period['STORE_NBR']==233
]['TOT_SALES']

t_stat, p_value = ttest_ind(
    trial_sales,
    control_sales,
    equal_var=False
)

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

T-statistic: 1.0590935477424306
P-value: 0.3389336098077161

```
from scipy.stats import ttest_ind

trial_store = 77
control_stores = [86, 88]

trial_period = store_monthly[
    store_monthly['YEARMONTH'] >= '201902'
]

trial_sales = trial_period[
    trial_period['STORE_NBR'] == trial_store
]['TOT_SALES']

for control_store in control_stores:

    control_sales = trial_period[
        trial_period['STORE_NBR'] == control_store
    ]['TOT_SALES']

    t_stat, p_value = ttest_ind(
        trial_sales,
        control_sales,
        equal_var=False
    )

    print(f"Store {trial_store} vs Store {control_store}")
    print(f"T-statistic: {t_stat:.4f}")
    print(f"P-value: {p_value:.4f}")
    print("-" * 30)
```

Store 77 vs Store 86
T-statistic: -17.9376
P-value: 0.0000

Store 77 vs Store 88
T-statistic: -34.9668
P-value: 0.0000
